

Note 4: The computation and decomposition of DLA

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4.1 Summary of Progress

This weekly note focuses on the computation and decomposition of dynamical Lie algebras (DLA).

1. We establish the structure theory of DLA inside $\mathfrak{u}(N)$.
2. We develop numerical decomposition algorithms with complexity analysis.
3. We propose efficient symbolic algorithms for Pauli-based DLA.
4. We test examples numerically and present conjectures.

4.2 Recall the work of the last week

Given $\mathcal{G} = \{H_1, \dots, H_k\} \subset \text{Herm}(N)$ and there are distinct levels of computing the DLA $\mathfrak{g} = \langle i\mathcal{G} \rangle_{\text{Lie}}$.

- **DLA Basis / Closure Computation.** Finds a real basis $\{Y_1, \dots, Y_d\} \subset \mathfrak{u}(N)$ such that

$$\mathfrak{g} = \text{span}_{\mathbb{R}}\{Y_1, \dots, Y_d\}.$$

- **DLA Structure Identification.** Determines the Lie-algebra isomorphism class of $\mathfrak{g}$, including its decomposition into center and simple summands:

$$\mathfrak{g} = \mathfrak{g}_1 \oplus \dots \oplus \mathfrak{g}_{k-1} \oplus \mathfrak{g}_k.$$

where $\mathfrak{g}_k$ is the center.

- **g-Purity.** Let $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ and $\{B_j\}_{j=1}^{\dim(\mathfrak{g})}$ be an orthonormal basis of $\mathfrak{g}_{\mathbb{C}}$ under the inner product $\langle A, B \rangle = \text{Tr}[A^\dagger B]$. $H_{\mathfrak{g}}$ denotes the orthogonal projection of H onto $\mathfrak{g}_{\mathbb{C}}$. Define

$$\mathcal{P}_{\mathfrak{g}}(H) = \text{Tr}[H_{\mathfrak{g}}^2] = \sum_{j=1}^{\dim(\mathfrak{g})} \left| \text{Tr}[B_j^\dagger H] \right|^2.$$

Note that The variance of $\ell_{\theta}(\rho, O)$ is

$$\text{Var}_{\theta}[\ell_{\theta}(\rho, O)] = \sum_{j=1}^{k-1} \frac{\mathcal{P}_{\mathfrak{g}_j}(\rho) \mathcal{P}_{\mathfrak{g}_j}(O)}{\dim(\mathfrak{g}_j)}.$$

We further distinguish two computational paradigms:

- *Symbolic DLA computation*
- *Numerical DLA computation*

Algorithm 1 Computable Inductive Closure Algorithm for DLA**Require:** Skew-Hermitian generators $\{X_1, X_2, \dots, X_k\} \subset \mathfrak{u}(N)$ **Ensure:** Real linearly independent basis $\mathcal{B}$ of $\mathfrak{g}_{\text{DLA}}$

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1:  $\mathcal{B} \leftarrow$  real-linearly independent subset of  $\{X_1, \dots, X_k\}$ 
2: while True do
3:    $\mathcal{C} \leftarrow \{AB - BA \mid A, B \in \mathcal{B}\}$ 
4:    $\mathcal{T} \leftarrow \mathcal{B} \cup \mathcal{C}$ 
5:    $\mathcal{B}_{\text{new}} \leftarrow$  real-linearly independent subset of  $\mathcal{T}$ 
6:   if  $|\mathcal{B}_{\text{new}}| = |\mathcal{B}|$  then
7:      $\mathcal{B} \leftarrow \mathcal{B}_{\text{new}}$ 
8:     break
9:   end if
10:   $\mathcal{B} \leftarrow \mathcal{B}_{\text{new}}$ 
11: end while
12: return  $\mathcal{B}$ 

```

Time complexity: $O(MN^5)$, where M is the number of closure iterations.

Algorithm 2 Computable Algorithm for $\mathfrak{g}$ -Purity**Require:** Real linearly independent basis $\mathcal{B} = \{Y_1, \dots, Y_d\}$ of $\mathfrak{g} \subset \mathfrak{u}(N)$ **Require:** Hermitian operator $H \in \text{Herm}(N)$ **Ensure:** $\mathfrak{g}$ -purity $\mathcal{P}_{\mathfrak{g}}(H) \in \mathbb{R}_{\geq 0}$

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1:  $\mathcal{B}_{\text{ortho}} \leftarrow$  Schmidt orthonormalize  $\mathcal{B}$  under  $\langle A, B \rangle = \text{Tr}(A^\dagger B)$ 
2:  $\mathcal{P} \leftarrow 0$ 
3: for  $B_j \in \mathcal{B}_{\text{ortho}}$  do
4:    $c_j \leftarrow \text{Tr}(B_j^\dagger H)$ 
5:    $\mathcal{P} \leftarrow \mathcal{P} + |c_j|^2$ 
6: end for
7: return  $\mathcal{P}$ 

```

4.2.1 Numerical computation For general Matrix**Algorithm 3** Numerical DLA Computation Flow

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1: Input: Hermitian generator set  $\mathcal{G}$ 
2: 1. Skew-Hermitian conversion:  $X \leftarrow iX$  for all  $X \in \mathcal{G}$ 
3: 2. Tensor product expansion: Build multi-qubit operators
4: 3. Embed operators to real vectors (real+imag parts)
5: 4. Filter initial independent basis via QR factorization
6: 5. Loop:
7:   a. Compute all pairwise commutators of current basis
8:   b. Merge commutators into operator set
9:   c. Refresh independent basis (QR factorization)
10:  d. Break if basis dimension unchanged
11: 6. Output: Dimension of  $\mathfrak{g}_{\text{DLA}}$ 

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Complexity $\sim O(N^5) = O(2^{5n}) = O(32^n)$

4.2.2 Symbolic DLA computation for Pauli Matrix

Definition 1 (Pauli Group) For n qubits system, we define the Pauli group by

$$\mathcal{P}_n := \left\{ i^k X^{a_1} Z^{b_1} \otimes \dots \otimes X^{a_n} Z^{b_n} \mid k \in \mathbb{Z}; a_j, b_j \in \{0, 1\} \right\} \simeq \mathbb{Z}_4 \ltimes_o \mathbb{F}_2^{2n}$$

Let $v = (a_1, b_1, \dots, a_n, b_n), w = (c_1, d_1, \dots, c_n, d_n)$.

Define the map $P : \mathbb{F}_2^{2n} \rightarrow \mathcal{P}_n$ and $\gamma : \mathbb{F}_2^{2n} \times \mathbb{F}_2^{2n} \rightarrow \mathbb{F}_2$ by

$$P(v) := X^{a_1} Z^{b_1} \otimes \dots \otimes X^{a_n} Z^{b_n}$$

$$\gamma(v, w) := b_1 c_1 + \dots + b_n c_n = v^T \Gamma w, \quad \Gamma = \bigoplus_{j=1}^n \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then we have

$$i^k P(v) i^l P(w) = i^{k+l} (-1)^{\gamma(v,w)} P(v+w)$$

Define the map $B : \mathbb{F}_2^{2n} \rightarrow \mathcal{P}_n$ and $\omega : \mathbb{F}_2^{2n} \times \mathbb{F}_2^{2n} \rightarrow \mathbb{F}_2$ by

$$B(v) := i^{1+\gamma(v,v)} P(v);$$

$$\omega(v, w) := \gamma(v, w) + \gamma(w, v) = v^T \Omega w, \quad \Omega = \bigoplus_{j=1}^n \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Then we have $\{B(v) \mid v \in \mathbb{F}_2^{2n}\}$ form a basis of $\mathfrak{u}(2^n)$, and the commutator given by

$$[B(v), B(w)] = \begin{cases} \pm 2B(v+w) & \omega(v, w) = 1. \\ 0 & \omega(v, w) = 0. \end{cases}$$

Algorithm 4 Vector-Encoded Inductive Closure Algorithm for DLA

Require: Skew-Hermitian generator basis $\{B_1, B_2, \dots, B_k\} \subset \mathfrak{u}(N)$

Ensure: Closed real basis $\mathcal{B}$ of the dynamical Lie algebra $\mathfrak{g}_{\text{DLA}}$

- 1: **Vector Encoding:** Map each B_i to its unique index vector $v_i \in \mathbb{F}_2^{2n}$,
 - 2: and identify $B_i \leftrightarrow B(v_i)$
 - 3: $\mathcal{B} \leftarrow$ distinct element subset of $\{B_1, B_2, \dots, B_k\}$
 - 4: **while** True **do**
 - 5: $\mathcal{C} \leftarrow \{B(v+w) \mid B(v), B(w) \in \mathcal{B}, \omega(v, w) \equiv 1 \pmod{2}\}$
 - 6: $\mathcal{T} \leftarrow \mathcal{B} \cup \mathcal{C}$
 - 7: $\mathcal{B}_{\text{new}} \leftarrow$ distinct element subset of $\mathcal{T}$
 - 8: **if** $|\mathcal{B}_{\text{new}}| = |\mathcal{B}|$ **then**
 - 9: $\mathcal{B} \leftarrow \mathcal{B}_{\text{new}}$
 - 10: **break**
 - 11: **end if**
 - 12: $\mathcal{B} \leftarrow \mathcal{B}_{\text{new}}$
 - 13: **end while**
 - 14: **return** $\mathcal{B}$
-

Remain Problem: Given a compact real semisimple Lie algebra $\mathfrak{g} \subseteq \mathfrak{u}(N)$, decompose $\mathfrak{g}$ into a direct sum of **simple ideals** and compute the dimension of each simple summand $\dim \mathfrak{g}_j$.

4.3 The structure and decomposition of DLA

4.3.1 The structure of DLA

We can define Lie bracket on complexification $\mathfrak{u}(N) \otimes_{\mathbb{R}} \mathbb{C}$ by

$$(A \otimes z)(B \otimes w) = AB \otimes zw \Rightarrow [A \otimes z, B \otimes w] := [A, B] \otimes zw$$

and we have proved that

Theorem 2 *There exists an isomorphism of complex Lie algebras:*

$$\mathfrak{u}(N) \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{gl}(N, \mathbb{C})$$

with the map $\varphi \left(\sum_{j=1}^t A_j \otimes z_j \right) = \sum_{j=1}^t z_j A_j$.

The isomorphism of complex Lie algebras has inverse map

$$\psi = \varphi^{-1} : \mathfrak{gl}(N, \mathbb{C}) \longrightarrow \mathfrak{u}(N) \otimes_{\mathbb{R}} \mathbb{C}$$

given explicitly by

$$\psi(X) = \frac{X - X^\dagger}{2} \otimes 1 + \frac{X + X^\dagger}{2i} \otimes i.$$

Define two real linear projection maps

$$P_1(X) = \frac{X - X^\dagger}{2}, \quad P_2(X) = \frac{X + X^\dagger}{2i},$$

which project $\mathfrak{gl}(N, \mathbb{C})$ onto $\mathfrak{u}(N)$. The inverse isomorphism is

$$\psi(X) = P_1(X) \otimes 1 + P_2(X) \otimes i.$$

Definition 3 A real Lie algebra $\mathfrak{h}$ is **compact** if there exists a negative-definite symmetric invariant bilinear form $B: \mathfrak{h} \times \mathfrak{h} \rightarrow \mathbb{R}$ such that

$$B([X, Y], Z) = B(X, [Y, Z])$$

for all $X, Y, Z \in \mathfrak{h}$.

Definition 4 Let $\mathfrak{g}$ be a real Lie algebra.

1. A subspace $\mathfrak{J} \subseteq \mathfrak{g}$ is an **ideal** if $[\mathfrak{J}, \mathfrak{g}] \subseteq \mathfrak{J}$.
2. The **center** of $\mathfrak{g}$ is the ideal

$$\mathfrak{z}(\mathfrak{g}) = \{X \in \mathfrak{g} \mid [X, Y] = 0, \forall Y \in \mathfrak{g}\}.$$

Definition 5 A real Lie algebra is **semisimple** if it has no nonzero solvable ideal. A real Lie algebra is **simple** if it is non-abelian and has no nontrivial ideals.

Lemma 6 Any subalgebra $\mathfrak{g} \subseteq \mathfrak{u}(N)$ is a compact real Lie algebra.

Proof Define the bilinear form on $\mathfrak{u}(N)$ by

$$B(X, Y) = -\operatorname{Re}(\operatorname{tr}(X^\dagger Y)).$$

We first verify **invariance**:

$$B([X, Y], Z) = -\operatorname{Re}(\operatorname{tr}((XY - YX)^\dagger Z)) = -\operatorname{Re}(\operatorname{tr}(Y^\dagger X^\dagger Z - X^\dagger Y^\dagger Z)).$$

Using cyclic property of trace $\operatorname{tr}(AB) = \operatorname{tr}(BA)$:

$$\operatorname{tr}(Y^\dagger X^\dagger Z) = \operatorname{tr}(X^\dagger Z Y^\dagger), \quad \operatorname{tr}(X^\dagger Y^\dagger Z) = \operatorname{tr}(Y^\dagger Z X^\dagger).$$

Thus

$$B([X, Y], Z) = -\operatorname{Re}(\operatorname{tr}(X^\dagger Z Y^\dagger - Y^\dagger Z X^\dagger)) = -\operatorname{Re}(\operatorname{tr}(X^\dagger(ZY^\dagger - Y^\dagger Z))) = B(X, [Y, Z]).$$

So B is invariant.

Next we verify **negative-definiteness**. For any $X \in \mathfrak{u}(N)$, we have $X^\dagger = -X$. Then

$$B(X, X) = -\operatorname{Re}(\operatorname{tr}(X^\dagger X)) = -\operatorname{Re}(\operatorname{tr}(-X^2)) = \operatorname{Re}(\operatorname{tr}(X^2)).$$

Since $X_{ji} = -\overline{X_{ij}}$,

$$\operatorname{tr}(X^2) = \sum_{i,j} X_{ij} X_{ji} = -\sum_{i,j} |X_{ij}|^2.$$

Hence

$$B(X, X) = -\sum_{i,j} |X_{ij}|^2 < 0 \quad \text{for } X \neq 0.$$

Thus B is negative-definite and invariant. The restriction to any subalgebra $\mathfrak{g} \subseteq \mathfrak{u}(N)$ preserves these properties, so $\mathfrak{g}$ is compact. ■

Lemma 7 Let $\mathfrak{h}$ be a compact real Lie algebra and $\mathfrak{J} \subseteq \mathfrak{h}$ an ideal. Then its orthogonal complement $\mathfrak{J}^\perp$ is also an ideal, and

$$\mathfrak{h} = \mathfrak{J} \oplus \mathfrak{J}^\perp$$

as vector spaces.

Proof The set $\mathfrak{J}^\perp$ is a subspace by linearity. For any $X \in \mathfrak{J}^\perp$, $Y \in \mathfrak{h}$, $Z \in \mathfrak{J}$, invariance of B gives

$$B([X, Y], Z) = B(X, [Y, Z]).$$

Since $\mathfrak{J}$ is an ideal, $[Y, Z] \in \mathfrak{J}$, so $B(X, [Y, Z]) = 0$. Hence $[X, Y] \in \mathfrak{J}^\perp$, so $\mathfrak{J}^\perp$ is an ideal. As B is non-degenerate, $\mathfrak{h} = \mathfrak{J} \oplus \mathfrak{J}^\perp$. ■

Lemma 8 Every semisimple compact real Lie algebra is a direct sum of simple compact real Lie algebras.

Proof We proceed by induction on $\dim_{\mathbb{R}} \mathfrak{s}$. If $\mathfrak{s}$ is simple, the claim holds. Otherwise, let $\mathfrak{g}_1$ be a proper ideal. By Lemma 7,

$$\mathfrak{s} = \mathfrak{g}_1 \oplus \mathfrak{g}_1^\perp, \quad [\mathfrak{g}_1, \mathfrak{g}_1^\perp] = \{0\}.$$

Both $\mathfrak{g}_1$ and $\mathfrak{g}_1^\perp$ are semisimple, compact, and of lower dimension. By the induction hypothesis, they decompose into simple compact summands. Hence $\mathfrak{s}$ does as well. ■

Theorem 9 Let $\mathfrak{g} \subseteq \mathfrak{u}(N)$ be any real subalgebra. Then $\mathfrak{g}$ admits a Lie algebra direct sum decomposition

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{g}_1 \oplus \mathfrak{g}_2 \oplus \cdots \oplus \mathfrak{g}_k,$$

where $\mathfrak{z}(\mathfrak{g})$ is the center of $\mathfrak{g}$, each $\mathfrak{g}_i$ is simple compact real, and

$$[\mathfrak{z}(\mathfrak{g}), \mathfrak{g}_i] = \{0\}, \quad [\mathfrak{g}_i, \mathfrak{g}_j] = \{0\} \quad (i \neq j).$$

Proof By Lemma 6, $\mathfrak{g}$ is compact. The center $\mathfrak{z}(\mathfrak{g})$ is an ideal. Let $\mathfrak{s} = \mathfrak{z}(\mathfrak{g})^\perp$. By Lemma 7,

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{s}, \quad [\mathfrak{z}(\mathfrak{g}), \mathfrak{s}] = \{0\}.$$

The summand $\mathfrak{s}$ is semisimple. By Lemma 8, $\mathfrak{s}$ decomposes into simple compact ideals

$$\mathfrak{s} = \mathfrak{g}_1 \oplus \cdots \oplus \mathfrak{g}_k$$

with $[\mathfrak{g}_i, \mathfrak{g}_j] = \{0\}$ for $i \neq j$. Combining the decompositions yields the result. ■

4.3.2 algorithm for decomposition

Lemma 10 Let $\mathfrak{g} = \text{span}\{X_1, X_2, \dots, X_d\} \subseteq \mathfrak{u}(N)$ be a real subalgebra. Then the center satisfies

$$\mathfrak{z}(\mathfrak{g}) = \bigcap_{i=1}^d \ker(\text{ad}_{X_i}).$$

Proof By definition,

$$\mathfrak{z}(\mathfrak{g}) = \{Z \in \mathfrak{g} \mid [Z, X] = 0, \forall X \in \mathfrak{g}\}.$$

Since $\text{ad}_X(Z) = [Z, X]$, we have

$$\mathfrak{z}(\mathfrak{g}) = \bigcap_{X \in \mathfrak{g}} \ker(\text{ad}_X) \subseteq \bigcap_{i=1}^d \ker(\text{ad}_{X_i}).$$

For the reverse inclusion, let

$$Z \in \bigcap_{i=1}^d \ker(\text{ad}_{X_i}), \quad \text{so} \quad [Z, X_i] = 0, \forall i.$$

Any $X \in \mathfrak{g}$ can be written as

$$X = \sum_{i=1}^d a_i X_i, \quad a_i \in \mathbb{R}.$$

By bilinearity of the Lie bracket,

$$[Z, X] = \sum_{i=1}^d a_i [Z, X_i] = 0.$$

Thus $Z \in \mathfrak{z}(\mathfrak{g})$. Hence

$$\mathfrak{z}(\mathfrak{g}) = \bigcap_{i=1}^d \ker(\text{ad}_{X_i}).$$

The intersection is finite and involves only the basis elements. ■

Algorithm 5 Numerical Algorithm for Computing the Center $\mathfrak{z}(\mathfrak{g})$ **Require:** Basis of Lie algebra $\mathcal{B} = \{X_1, X_2, \dots, X_d\} \subset \mathfrak{g} \subseteq \mathfrak{u}(N)$ **Ensure:** Basis of center $\mathfrak{z}(\mathfrak{g})$

- 1: Let any element $Z \in \mathfrak{g}$ be expressed as $Z = \sum_{t=1}^d \lambda_t X_t$, where $\lambda = (\lambda_1, \dots, \lambda_d)^\top \in \mathbb{R}^d$ are unknowns.
- 2: **for** $i \leftarrow 1$ **to** d **do**
- 3: Expand bracket condition $[X_i, Z] = \mathbf{0}$
- 4: Substitute linear combination of Z , rearrange into linear equations: $\sum_{t=1}^d [X_i, X_t] \lambda_t = \mathbf{0}$
- 5: **end for**
- 6: Assemble all above equations into a global homogeneous linear system: $A\lambda = \mathbf{0}$
- 7: Solve the system, compute the solution null space
- 8: Reconstruct center basis from null-space coefficients λ
- 9: **return** Basis of $\mathfrak{z}(\mathfrak{g})$

Time complexity: $O(16^n)$, for $N = 2^n$.**Algorithm 6** Numerical Computation of Orthogonal Complement**Require:** Ambient space basis $\mathcal{U} = \{U_1, U_2, \dots, U_m\}$, Subspace basis $\mathcal{V} = \{V_1, V_2, \dots, V_s\}$, Inner product $B(X, Y) = -\text{Re}(\text{tr}(X^\dagger Y))$ **Ensure:** Basis of orthogonal complement $\mathcal{V}^\perp = \{Y \in \text{span}(\mathcal{U}) \mid B(Y, V) = 0, \forall V \in \mathcal{V}\}$

- 1: Let arbitrary element $Y \in \text{span}(\mathcal{U})$ be written as $Y = \sum_{k=1}^m \mu_k U_k$, where $\mu = (\mu_1, \dots, \mu_m)^\top \in \mathbb{R}^m$ are unknowns.
- 2: **for** $j \leftarrow 1$ **to** s **do**
- 3: Establish orthogonal constraint: $B(Y, V_j) = 0$
- 4: Substitute expression of Y , obtain linear equation in μ
- 5: **end for**
- 6: Assemble all constraints into a homogeneous linear system $C\mu = \mathbf{0}$
- 7: Compute the nullspace of C
- 8: Construct orthogonal complement basis from nullspace coefficient vectors
- 9: **return** Basis of $\mathcal{V}^\perp$

Lemma 11 Let $\mathfrak{s}$ be a finite-dimensional compact semisimple real Lie algebra with basis $\{U_1, U_2, \dots, U_m\}$. For any non-zero element $X \in \mathfrak{s}$, define the subspace

$$\mathfrak{j}_X := \text{span} \left\{ \text{ad}_{U_{i_1}} \text{ad}_{U_{i_2}} \cdots \text{ad}_{U_{i_k}}(X) \mid k \geq 0, 1 \leq i_1, \dots, i_k \leq m \right\},$$

where $\text{ad}_U(V) = [U, V]$. Then:

1. $\mathfrak{j}_X$ is a non-zero ideal of $\mathfrak{s}$;
2. If $\dim \mathfrak{j}_X < \dim \mathfrak{s}$, then $\mathfrak{j}_X$ is a proper non-trivial ideal;
3. If $\mathfrak{s}$ is **simple**, then $\mathfrak{j}_X = \mathfrak{s}$ for all non-zero $X \in \mathfrak{s}$.

Proof (1) By definition $\mathfrak{j}_X \ni X$, so $\mathfrak{j}_X \neq \{0\}$.

Take any $U \in \{U_1, \dots, U_m\}$ and any generator $\text{ad}_{U_{i_1}} \cdots \text{ad}_{U_{i_k}}(X) \in \mathfrak{j}_X$:

$$\text{ad}_U(\text{ad}_{U_{i_1}} \cdots \text{ad}_{U_{i_k}}(X)) = \text{ad}_U \text{ad}_{U_{i_1}} \cdots \text{ad}_{U_{i_k}}(X),$$

which also belongs to the spanning set of j_X . Thus $\text{ad}_U(j_X) \subseteq j_X$ for all basis elements U . Since $\text{ad}_\cdot$ is linear, $\text{ad}_Y(j_X) \subseteq j_X$ holds for all $Y \in \mathfrak{s}$. Hence j_X is an ideal.

(2) Trivial by dimension comparison.

(3) If $\mathfrak{s}$ is simple, it has no proper non-trivial ideals. Since j_X is a non-zero ideal, necessarily $j_X = \mathfrak{s}$. ■

Therefore, a proper ideal can be constructed by: 1. picking a nonzero $X \in \mathfrak{s}$; 2. closing X under all adjoint actions of the basis; 3. checking its dimension.

Algorithm 7 Find a Proper Ideal of a Compact Semisimple Lie Algebra

Require: Basis $\mathcal{U} = \{U_1, \dots, U_m\}$ of compact semisimple $\mathfrak{i}$

Ensure: Proper non-trivial ideal $j \subsetneq \mathfrak{i}$, or **nil** if $\mathfrak{i}$ is simple

```

1: Pick any non-zero element  $X \in \mathfrak{i}$ 
2: Initialize subspace  $j \leftarrow \text{span}\{X\}$ 
3: repeat
4:   Initialize  $\mathcal{W} \leftarrow \emptyset$ 
5:   for each  $Y \in \text{basis of } j$  do
6:     for each  $U_k \in \mathcal{U}$  do
7:       Compute  $W = [U_k, Y] = \text{ad}_{U_k}(Y)$ 
8:       Add  $W$  to  $\mathcal{W}$ 
9:     end for
10:  end for
11:  Update  $j \leftarrow \text{span}(\{\text{basis of } j\} \cup \mathcal{W})$ 
12: until  $j$  is invariant and no new vectors appear
13: if  $\dim(j) < \dim(\mathfrak{i})$  then
14:   return  $j$ 
15: else
16:   return nil   {  $\mathfrak{i}$  is simple }
17: end if
```

Time complexity: $O(16^n)$.

Algorithm 8 Decomposition of $\mathfrak{g} \subseteq \mathfrak{u}(N)$

Require: Basis $\mathcal{B} = \{X_1, \dots, X_d\}$ of $\mathfrak{g} \subseteq \mathfrak{u}(N)$

Ensure: Center $\mathfrak{z}$, simple ideals $\mathcal{S}$ such that $\mathfrak{g} = \mathfrak{z} \oplus \bigoplus_{\mathfrak{s} \in \mathcal{S}} \mathfrak{s}$

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1: Compute center  $\mathfrak{z}$  using Algorithm 5
2: Compute semisimple part  $\mathfrak{s}$  using Algorithm 6
3: Initialize queue  $\mathcal{Q} \leftarrow [\mathfrak{s}]$ ,  $\mathcal{S} \leftarrow \emptyset$ 
4: while  $\mathcal{Q}$  is not empty do
5:   Pop  $\mathfrak{i}$  from  $\mathcal{Q}$ 
6:   Find ideal  $j$  using Algorithm 7
7:   if  $j = \text{nil}$  then
8:      $\mathcal{S} \leftarrow \mathcal{S} \cup \{\mathfrak{i}\}$ 
9:   else
10:    Compute  $j^\perp$  using Algorithm 6
11:    Push  $j, j^\perp$  into  $\mathcal{Q}$ 
12:   end if
13: end while
14: return  $\mathfrak{z}, \mathcal{S}$ 
```

The algorithm is suitable for general Lie algebra $\mathfrak{g} \subseteq \mathfrak{u}(N)$. Based on the Lie algebra decomposition algorithms and lemmas established earlier, we specialize to the DLA generated by Pauli matrices. Unlike general numerical linear algebra operations, Pauli generators have a unique binary symplectic structure, allowing us to optimize the decomposition via symbolic finite-field arithmetic instead of high-dimensional matrix computations.

Lemma 12 Let $\mathfrak{g} = \text{span}\{B(v) \mid v \in V\}$ be a Lie algebra generated by Pauli basis elements, where $V \subseteq \mathbb{F}_2^{2n}$ and ω is the symplectic form. Then:

1. The center is $\mathfrak{z}(\mathfrak{g}) = \text{span}\{B(u) \mid \omega(u, v) = 0, \forall v \in V\}$.
2. Every ideal of $\mathfrak{g}$ corresponds to an ω -invariant subspace of V .
3. The adjoint action $\text{ad}_{B(v)}(B(w))$ reduces to binary vector addition $v + w$ over $\mathbb{F}_2^{2n}$.

Proof (1) By definition, $[B(u), B(v)] = 0$ if and only if $\omega(u, v) = 0$. By Lemma 10, the center consists of elements commuting with all generators, which gives the symplectic orthogonal subspace.

(2) A subspace $\mathfrak{j} \subseteq \mathfrak{g}$ is an ideal iff $[\mathfrak{g}, \mathfrak{j}] \subseteq \mathfrak{j}$. Since $[B(v), B(w)] \propto B(v + w)$, invariance under Lie brackets is equivalent to invariance under addition in $\mathbb{F}_2^{2n}$.

(3) Directly from the commutation relation: $[B(v), B(w)] \propto B(v + w)$ if $\omega(v, w) = 1$, and zero otherwise. Thus the adjoint action is fully encoded by binary symplectic operations. ■

Then we have the following algorithms:

Algorithm 9 Symbolic Center Computation for Pauli DLA

Require: Generators $V = \{v_1, \dots, v_d\} \subset \mathbb{F}_2^{2n}$

Ensure: Binary basis U of the center $\mathfrak{z}(\mathfrak{g})$

- 1: Build symplectic matrix $\Omega = \bigoplus_{j=1}^n \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$
 - 2: Form linear system over $\mathbb{F}_2$: $v^\top \Omega u = 0, \forall v \in V$
 - 3: Assemble matrix $M \in \mathbb{F}_2^{d \times 2n}$ with rows $v_i^\top \Omega$
 - 4: Compute binary nullspace of M to obtain U
 - 5: **return** U
-

Time complexity: $O(n^3)$, dominated by Gaussian elimination over the low-dimensional binary field $\mathbb{F}_2^{2n}$.

Algorithm 10 Symbolic Proper Ideal Search for Pauli DLA

Require: Subspace $V \subset \mathbb{F}_2^{2n}$

Ensure: Proper ideal subspace $J \subsetneq V$, or **nil** if simple

- 1: Pick nonzero $x \in V$
 - 2: Initialize $J \leftarrow \text{span}\{x\}$
 - 3: **repeat**
 - 4: Expand J with $v + w$ for all $v \in V, w \in J$
 - 5: Reduce to $\mathbb{F}_2$ -span
 - 6: **until** J is invariant
 - 7: **if** $\dim J < \dim V$ **then**
 - 8: **return** J
 - 9: **else**
 - 10: **return** **nil**
 - 11: **end if**
-

Time complexity: $O(n \cdot 2^n)$?

Algorithm 11 Symbolic Orthogonal Complement for Pauli DLA

Require: Subspace $W \subset \mathbb{F}_2^{2n}$, ambient space $\mathbb{F}_2^{2n}$

Ensure: Symplectic orthogonal complement $W^\perp$

- 1: Build linear constraints: $w^\top \Omega u = 0, \forall w \in W$
 - 2: Assemble system matrix over $\mathbb{F}_2$
 - 3: Compute nullspace to get $W^\perp$
 - 4: **return** $W^\perp$
-

Time complexity: $O(n^3)$, dominated by binary Gaussian elimination for symplectic constraint systems.

4.4 Numerical test and some conjectures

Case 1: $\mathcal{G} = \{X_i\} \cup \{Z_i Z_{i+1}\}$

Table 4.1: Numerical results for Case 1

n	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
$\dim \mathfrak{g}$	6	15	28	45	66	91	120	153	190	231	276	325	378	435	496	561	630	703	780

Conjecture 1 Let $\mathcal{G} = \{X_i\} \cup \{Z_i Z_{i+1}\}$, then

$$\dim \mathfrak{g} = n(2n - 1), \quad \mathfrak{g} \simeq \mathfrak{so}(2n).$$

Case 2: $\mathcal{G} = \{X_i\} \cup \{Z_i Z_j \mid i < j\}$

Table 4.2: Numerical results for Case 2

n	2	3	4	5	6	7
$\dim \mathfrak{g}$	6	30	126	510	2046	8190

Conjecture 2 Let $N = 2^n$ and $\mathcal{G} = \{X_i\} \cup \{Z_i Z_j \mid i < j\}$. Then

$$\dim \mathfrak{g} = 2^{2n-1} - 2.$$

Case 3: $\mathcal{G} = \{X_i\} \cup \{Z_1 Z_2 \cdots Z_n\}$

Table 4.3: Numerical results for Case 3

n	2	3	4	5	6	7	8	9	10	11
$\dim \mathfrak{g}$	6	12	24	48	96	192	384	768	1536	3072

Conjecture 3 Let $N = 2^n$ and $\mathcal{G} = \{X_i\} \cup \{Z_1 Z_2 \cdots Z_n\}$. Then

$$\dim \mathfrak{g} = 3 \cdot 2^{n-1}.$$

Problem: Determine the structure of case 2,3.

4.5 Plan for Next Week

4.6 Appendix

4.6.1 Solutions to exercises and problems

Definition 13 A subset $\mathcal{C} \subseteq \mathbb{F}_2^n$ is an $[n, k, d]$ -Binary Linear code if it satisfies

$$\mathcal{C} \leq \mathbb{F}_2^n, \quad \dim_{\mathbb{F}_2} \mathcal{C} = k, \quad d = \min_{c \neq c' \in \mathcal{C}} d_H(c, c').$$

Lemma 14 The ℓ -repetition code is an $[\ell, 1, \ell]$ -linear code over $\mathbb{F}_2$.

Consider the ℓ -repetition code.

1. Give the encoding function $\mathcal{E}$.
2. List all codewords.
3. Show that it is an $[\ell, 1, \ell]$ -linear code.
4. Compute its error-detecting and error-correcting capabilities.
5. Compute the rate R and $\lim_{\ell \rightarrow \infty} R$.

Proof

1. The encoding function is

$$\mathcal{E} : \mathbb{F}_2 \rightarrow \mathbb{F}_2^\ell, \quad \mathcal{E}(0) = (0, \dots, 0), \quad \mathcal{E}(1) = (1, \dots, 1).$$

2. The code has exactly two codewords:

$$\mathcal{C} = \{(0, \dots, 0), (1, \dots, 1)\}.$$

3. The code length is $n = \ell$, the dimension is $k = 1$, and the minimum Hamming distance is

$$d = \min_{c \neq c'} d_H(c, c') = \ell.$$

Thus $\mathcal{C}$ is an $[\ell, 1, \ell]$ -linear code.

4. The error-detecting capability is $d - 1 = \ell - 1$. The error-correcting capability is

$$\left\lfloor \frac{d-1}{2} \right\rfloor = \left\lfloor \frac{\ell-1}{2} \right\rfloor.$$

5. The rate is

$$R = \frac{k}{n} = \frac{1}{\ell}, \quad \lim_{\ell \rightarrow \infty} R = 0.$$

■

Let $\mathcal{C}$ be an $[n, k, d]$ -binary code. Prove that

1. $\mathcal{C}$ can detect up to $d - 1$ bit-flip errors.
2. $\mathcal{C}$ can correct up to $\left\lfloor \frac{d-1}{2} \right\rfloor$ bit-flip errors.

Proof

1. Let $c \in \mathcal{C}$ and $d_H(c, y) = t \leq d - 1$. For any $c' \in \mathcal{C}$ with $c' \neq c$, we have $d_H(c, c') \geq d$. By the triangle inequality

$$d_H(y, c') \geq d_H(c, c') - d_H(c, y) \geq d - t \geq 1.$$

Thus $y \notin \mathcal{C}$, so the error is detected.

2. Let $t = \lfloor \frac{d-1}{2} \rfloor$, so $2t + 1 \leq d$. Let $d_H(c, y) \leq t$. For any $c' \neq c$,

$$d_H(y, c') \geq d - t > t \geq d_H(c, y).$$

Thus c is the unique nearest codeword, so the error is corrected. ■

Let $\mathcal{C}$ be an $[n, k, d]$ -linear code over $\mathbb{F}_2$. Prove that

$$|\mathcal{C}| = 2^k.$$

Proof Since $\mathcal{C}$ is a k -dimensional vector space over $\mathbb{F}_2$, it has a basis $\{g_1, \dots, g_k\}$. Every codeword can be uniquely written as

$$c = \sum_{i=1}^k a_i g_i, \quad a_i \in \mathbb{F}_2.$$

Each a_i has two choices, so the total number of codewords is

$$|\mathcal{C}| = 2^k. \quad \text{■}$$

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